OLA DATASET Analysis

1. Project Overview

The **OLA Data Analyst Project** focuses on analyzing ride-booking data from OLA using **SQL** and **Power BI** to generate meaningful business insights. The project helps in understanding customer behavior, ride trends, booking performance, cancellations, payment preferences, revenue generation, and customer/driver satisfaction.

Using SQL queries and Power BI dashboards, the project provides interactive visualizations and analytical reports that support data-driven decision-making. Key insights include ride volume trends, booking status analysis, vehicle-wise performance, revenue by payment method, cancellation reasons, and ratings comparison between customers and drivers.

The dashboard is designed to give a complete operational overview of ride services and improve business efficiency through data analytics techniques.

2. Dataset Summary

The dataset used in this project contains ride-booking information collected from OLA operations. It includes details related to bookings, customers, vehicle types, ride distances, cancellations, payment methods, ratings, and booking values.

Main Data Fields:

- Booking ID
- Date & Time
- Customer ID
- Vehicle Type
- Pickup & Drop Location
- Booking Status
- Ride Distance
- Booking Value
- Payment Method
- Driver Ratings
- Customer Ratings
- Cancellation Reasons
- Incomplete Ride Reasons

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in PythoN:

- **Data Loading:** Imported the dataset using pandas.

- **Initial Exploration:** Used df.info() to check structure for summary statistics.

[2]:
```
import pandas as pd
df = pd.read_excel(r"C:\Users\hp\Downloads\ola_project.xlsx")
df.head()
```

[2]:

	Date	Time	Booking_ID	Booking_Status	Customer_ID	Vehicle_Type	Pickup_Location	Drop_Location	V_TAT	C_TAT	Canceled_Rides_by_Customer	Canceled
0	2024-07-26 14:00:00	14:00:00	CNR7153255142	Canceled by Driver	CID713523	Prime Sedan	Tumkur Road	RT Nagar	NaN	NaN	NaN	Pers
1	2024-07-25 22:20:00	22:20:00	CNR2940424040	Success	CID225428	Bike	Magadi Road	Varthur	203.0	30.0	NaN	
2	2024-07-30 19:59:00	19:59:00	CNR2982357879	Success	CID270156	Prime SUV	Sahakar Nagar	Varthur	238.0	130.0	NaN	
3	2024-07-22 03:15:00	03:15:00	CNR2395710036	Canceled by Customer	CID581320	eBike	HSR Layout	Vijayanagar	NaN	NaN	Driver is not moving towards pickup location	
4	2024-07-02 09:02:00	09:02:00	CNR1797421769	Success	CID939555	Mini	Rajajinagar	Chamarajpet	252.0	80.0	NaN	

- **Missing Data Handling:** Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.

- **Column Standardization:** Renamed columns to **snake case** for better readability and documentation.

- **Database Integration:** Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions:

1. **RETRIEVE ALL SUCCESSFUL BOOKINGS**

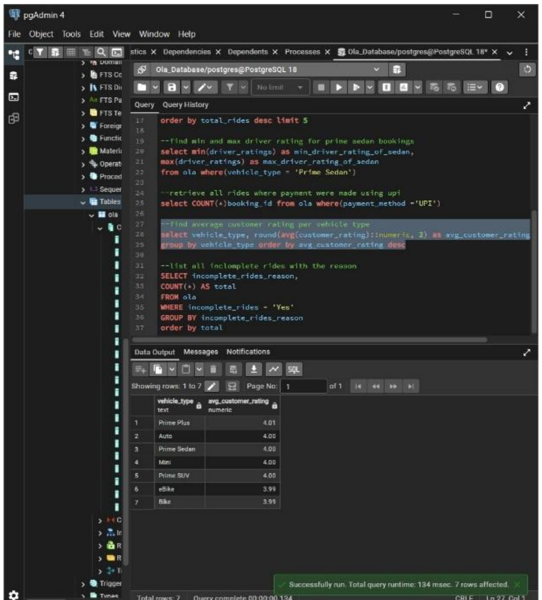
	booking_id text	booking_status text
1	CNR2940424040	Success
2	CNR2982357879	Success
3	CNR1797421769	Success
4	CNR8787177882	Success
5	CNR3612067560	Success
6	CNR4787583516	Success
7	CNR7943634301	Success
8	CNR4524472111	Success
9	CNR8181602032	Success
10	CNR8090918544	Success
11	CNR3196156650	Success
12	CNR9975925287	Success
13	CNR1591113431	Success

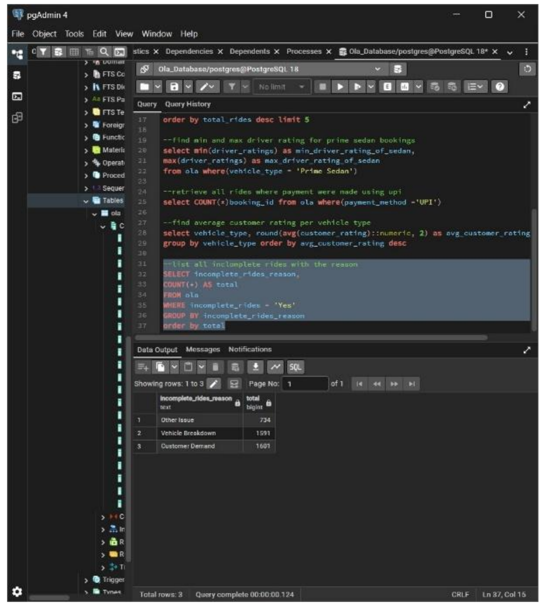
2. FIND AVG RIDE DISTANCE FOR EACH VEHICLE

	min_driver_rating_of_sedan double precision	max_driver_rating_of_sedan double precision
1	3	5

3. TOP 5 CUSTOMERS

	customer_id text	total_rides bigint
1	CID268274	4
2	CID635963	4
3	CID836942	4
4	CID266327	4
5	CID288207	4


pgAdmin 4
File Object Tools Edit View Window Help
Ola_Database/postgres@PostgreSQL 18
order by total_rides desc limit 5
--find min and max driver rating for prime sedan bookings
select min(driver_ratings) as min_driver_rating_of_sedan,
max(driver_ratings) as max_driver_rating_of_sedan
from ola where(vehicle_type = 'Prime Sedan')
--retrieve all rides where payment were made using upi
select COUNT(*)booking_id from ola where(payment_method ='UPI')
--find average customer rating per vehicle type
select vehicle_type, round(avg(customer_rating)::numeric, 2) as avg_customer_rating
group by vehicle_type order by avg_customer_rating desc
--list all inclomplete rides with the reason
SELECT incomplete_rides_reason,
COUNT(*) AS total
FROM ola
WHERE incomplete_rides = 'Yes'
GROUP BY incomplete_rides_reason
order by total
Data Output Messages Notifications
Showing rows: 1 to 7 Page No: 1 of 1
vehicle_type text | avg_customer_rating numeric
1 Prime Plus 4.01
2 Auto 4.00
3 Prime Sedan 4.00
4 Mini 4.00
5 Prime SUV 4.00
6 eBike 3.99
7 Bike 3.99
Successfully run. Total query runtime: 134 msec. 7 rows affected.
Total rows: 7 Query complete 00:00:00.134 CRLF Ln 27, Col 1


pgAdmin 4
File Object Tools Edit View Window Help
Ola_Database/postgres@PostgreSQL 18
order by total_rides desc limit 5
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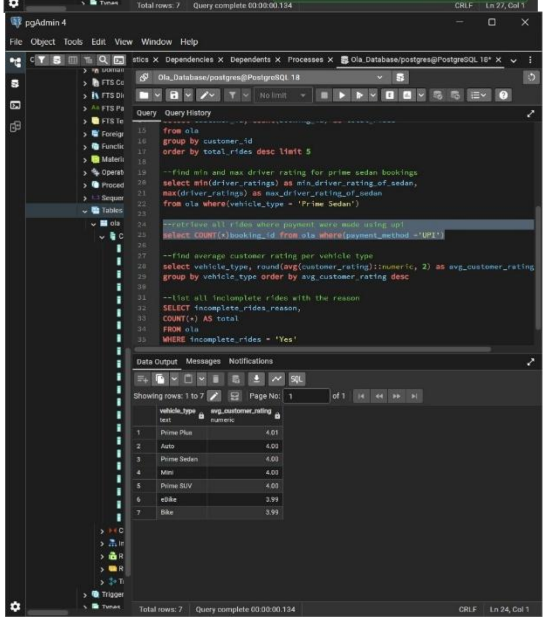
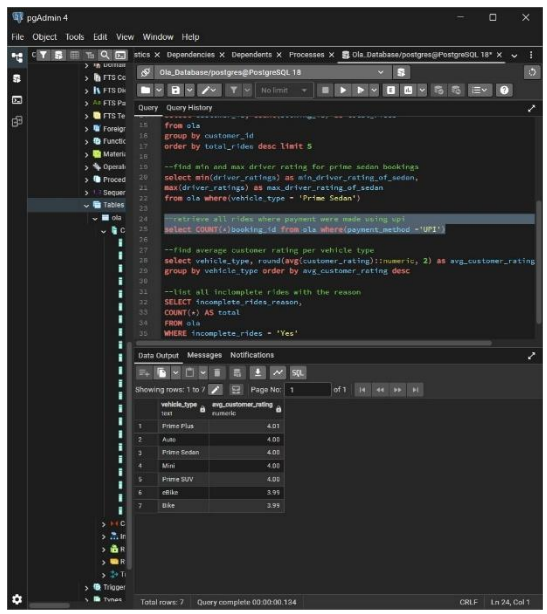
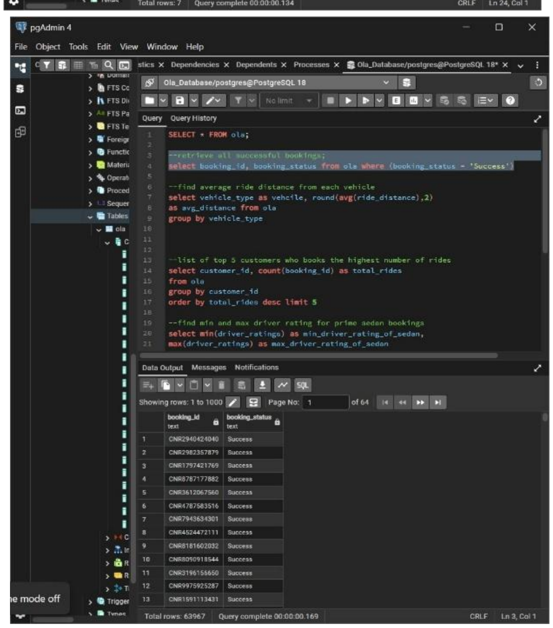
group by vehicle_type order by avg_customer_rating desc
--list all inclomplete rides with the reason
SELECT incomplete_rides_reason,
COUNT(*) AS total
FROM ola
WHERE incomplete_rides = 'Yes'
GROUP BY incomplete_rides_reason
order by total
Data Output Messages Notifications
Showing rows: 1 to 3 Page No: 1 of 1
incomplete_rides_reason text | total bigint
1 Other Issue 734
2 Vehicle Breakdown 1591
3 Customer Demand 1601
Total rows: 3 Query complete 00:00:00.134 CRLF Ln 37, Col 15

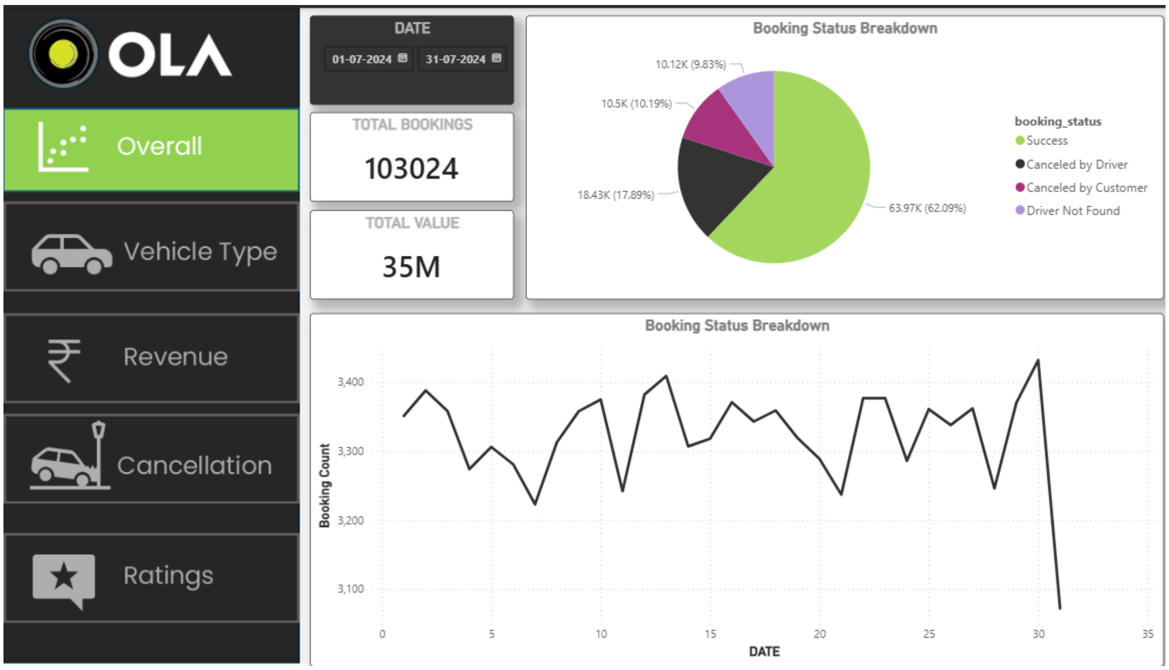

pgAdmin 4
File Object Tools Edit View Window Help
Ola_Database/postgres@PostgreSQL 18
from ola
group by customer_id
order by total_rides desc limit 5
--find min and max driver rating for prime sedan bookings
select min(driver_ratings) as min_driver_rating_of_sedan,
max(driver_ratings) as max_driver_rating_of_sedan
from ola where(vehicle_type = 'Prime Sedan')
--retrieve all rides where payment were made using upi
select COUNT(*)booking_id from ola where(payment_method ='UPI')
--find average customer rating per vehicle type
select vehicle_type, round(avg(customer_rating)::numeric, 2) as avg_customer_rating
group by vehicle_type order by avg_customer_rating desc
--list all inclomplete rides with the reason
SELECT incomplete_rides_reason,
COUNT(*) AS total
FROM ola
WHERE incomplete_rides = 'Yes'
Data Output Messages Notifications
Showing rows: 1 to 7 Page No: 1 of 1
vehicle_type text | avg_customer_rating numeric
1 Prime Plus 4.01
2 Auto 4.00
3 Prime Sedan 4.00
4 Mini 4.00
5 Prime SUV 4.00
6 eBike 3.99
7 Bike 3.99
Total rows: 7 Query complete 00:00:00.134 CRLF Ln 24, Col 1

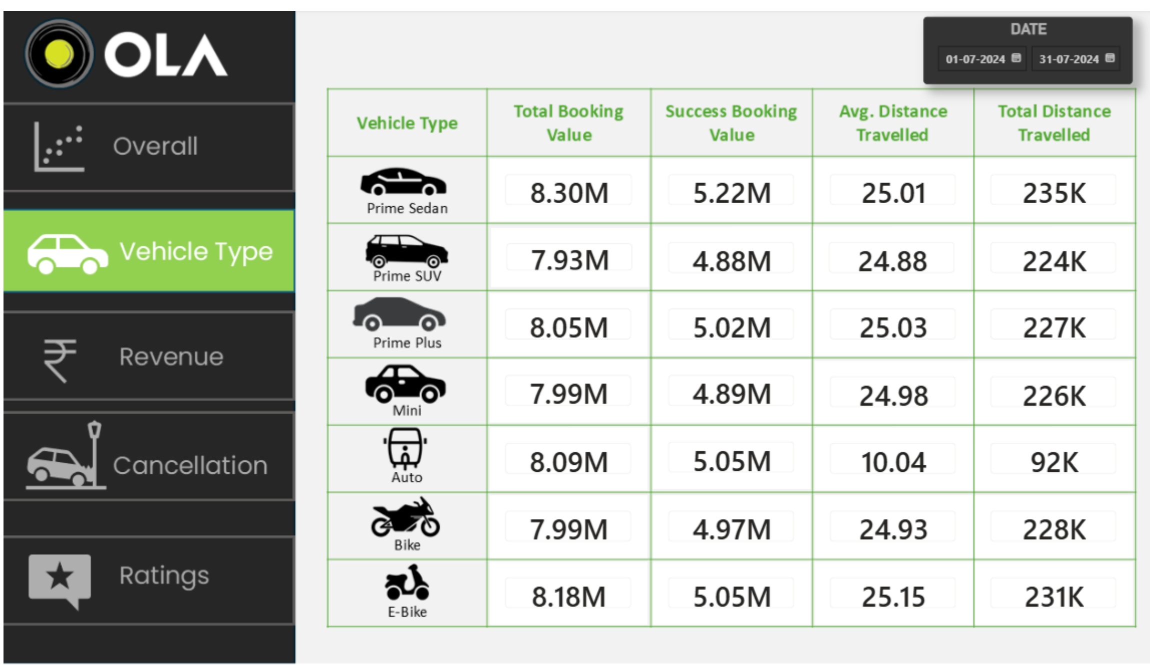

pgAdmin 4
File Object Tools Edit View Window Help
Ola_Database/postgres@PostgreSQL 18
from ola
group by customer_id
order by total_rides desc limit 5
--find min and max driver rating for prime sedan bookings
select min(driver_ratings) as min_driver_rating_of_sedan,
max(driver_ratings) as max_driver_rating_of_sedan
from ola where(vehicle_type = 'Prime Sedan')
--retrieve all rides where payment were made using upi
select COUNT(*)booking_id from ola where(payment_method ='UPI')
--find average customer rating per vehicle type
select vehicle_type, round(avg(customer_rating)::numeric, 2) as avg_customer_rating
group by vehicle_type order by avg_customer_rating desc
--list all inclomplete rides with the reason
SELECT incomplete_rides_reason,
COUNT(*) AS total
FROM ola
WHERE incomplete_rides = 'Yes'
Data Output Messages Notifications
Showing rows: 1 to 7 Page No: 1 of 1
vehicle_type text | avg_customer_rating numeric
1 Prime Plus 4.01
2 Auto 4.00
3 Prime Sedan 4.00
4 Mini 4.00
5 Prime SUV 4.00
6 eBike 3.99
7 Bike 3.99
Total rows: 7 Query complete 00:00:00.134 CRLF Ln 24, Col 1


pgAdmin 4
File Object Tools Edit View Window Help
Ola_Database/postgres@PostgreSQL 18
SELECT * FROM ola;
--retrieve all successful bookings;
select booking_id, booking_status from ola where (booking_status = 'Success')
--find average ride distance from each vehicle
select vehicle_type as vehcile, round(avg(ride_distance),2)
as avg_distance from ola
group by vehicle_type
--list of top 5 customers who books the highest number of rides
select customer_id, count(booking_id) as total_rides
from ola
group by customer_id
order by total_rides desc limit 5
--find min and max driver rating for prime sedan bookings
select min(driver_ratings) as min_driver_rating_of_sedan,
max(driver_ratings) as max_driver_rating_of_sedan
Data Output Messages Notifications
Showing rows: 1 to 1000 Page No: 1 of 64
booking_id text | booking_status text
1 CNR2940424040 Success
2 CNR2982357879 Success
3 CNR1797421769 Success
4 CNR8787177882 Success
5 CNR3612067560 Success
6 CNR4787583516 Success
7 CNR7943634301 Success
8 CNR4524472111 Success
9 CNR8181602032 Success
10 CNR8090918544 Success
11 CNR3196156650 Success
12 CNR9975925287 Success
13 CNR1591113431 Success
Total rows: 63967 Query complete 00:00:00.169 CRLF Ln 3, Col 1

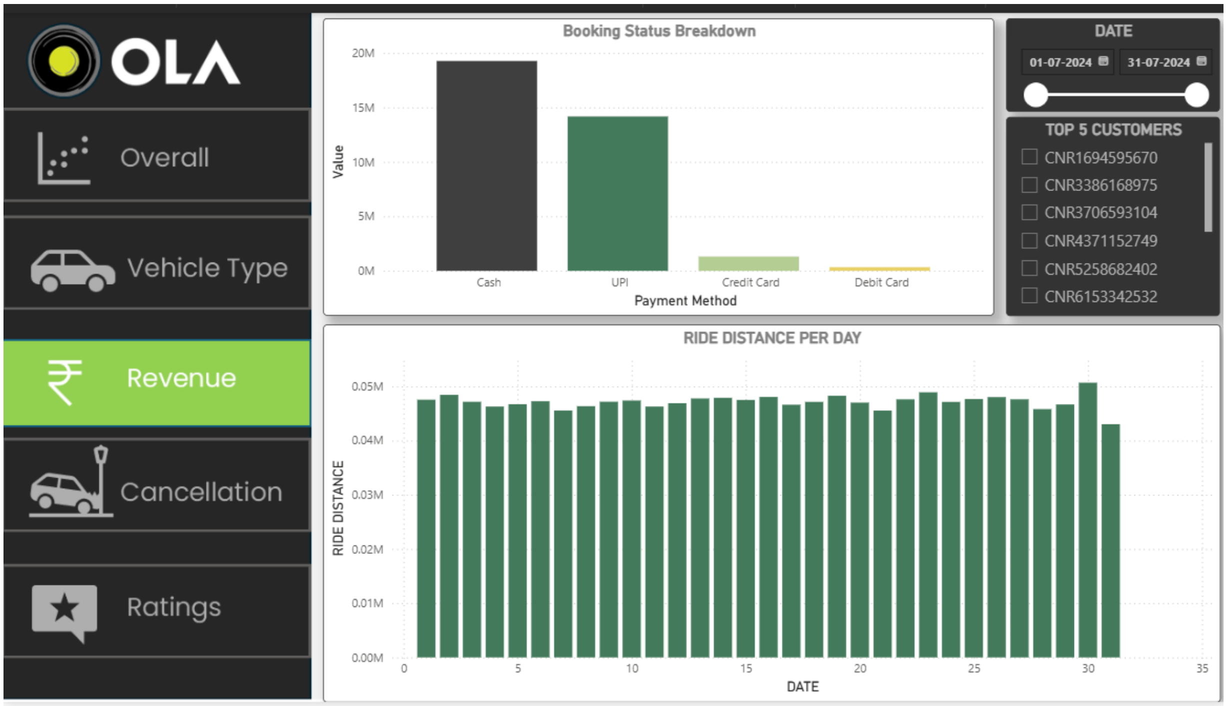
5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.

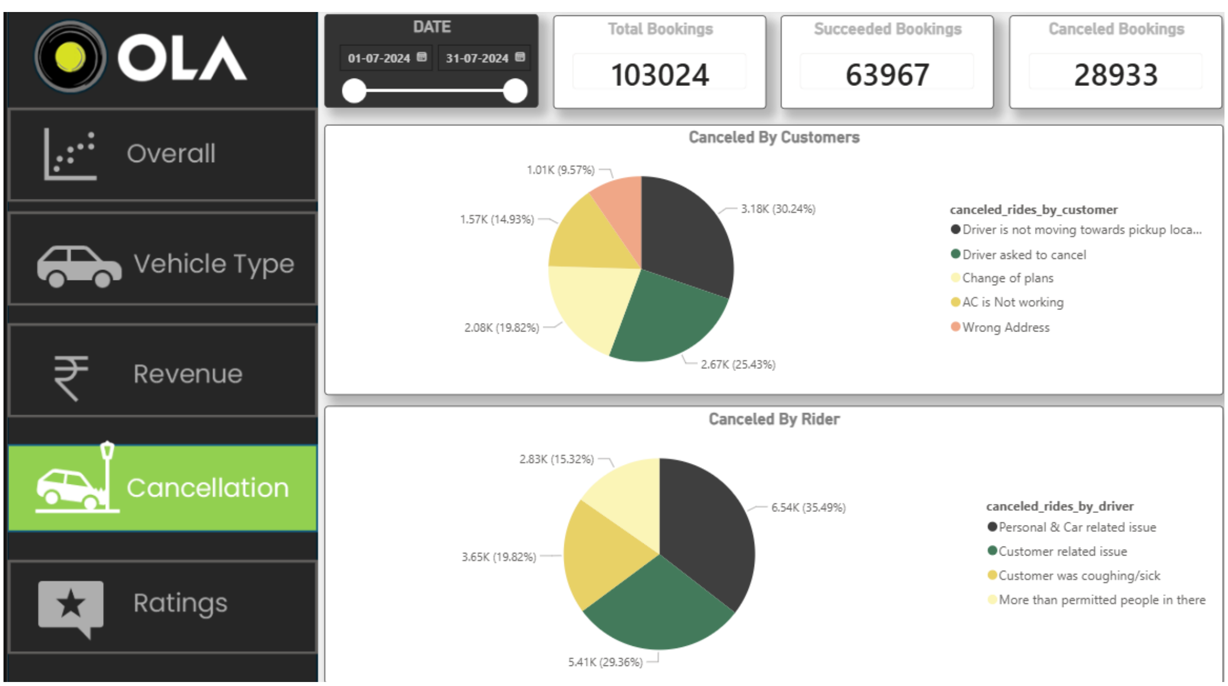




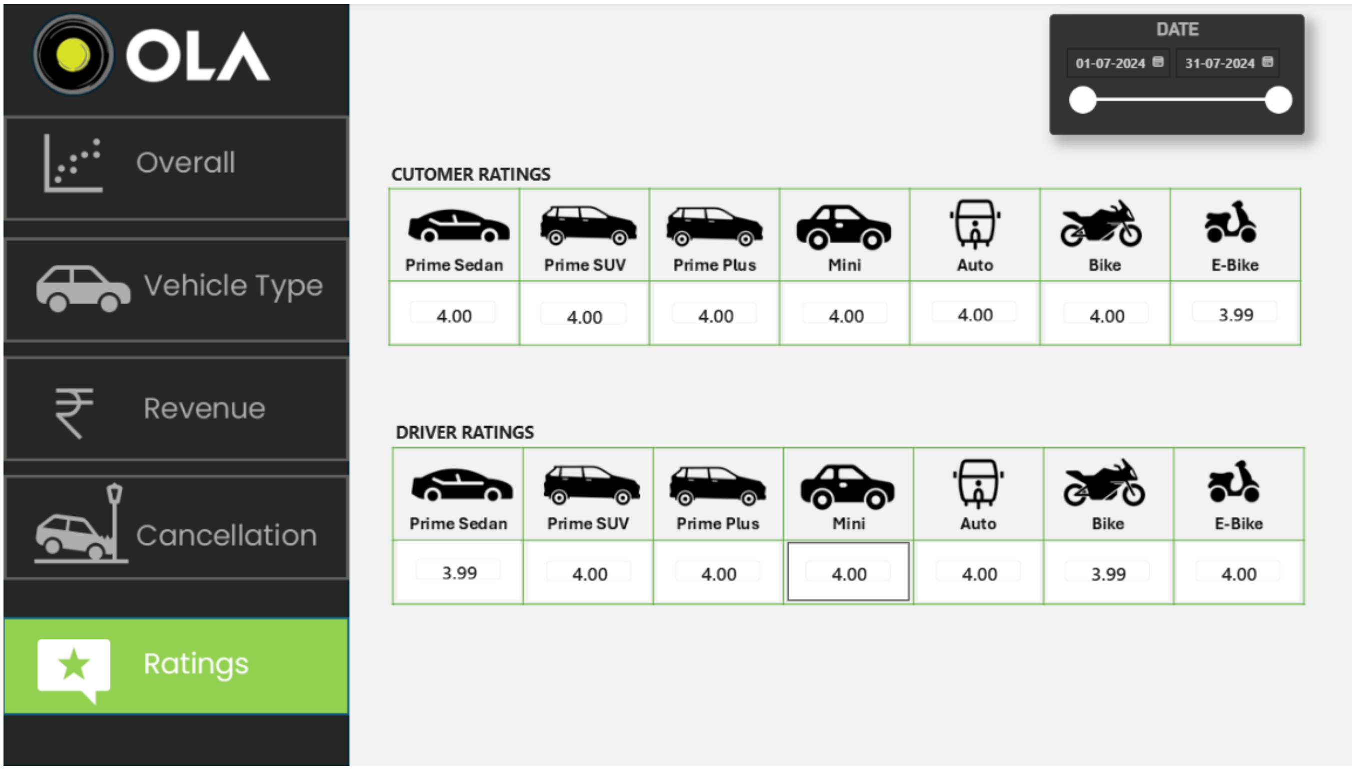
Vehicle Type	Total Booking Value	Success Booking Value	Avg. Distance Travelled	Total Distance Travelled
Prime Sedan	8.30M	5.22M	25.01	235K
Prime SUV	7.93M	4.88M	24.88	224K
Prime Plus	8.05M	5.02M	25.03	227K
Mini	7.99M	4.89M	24.98	226K
Auto	8.09M	5.05M	10.04	92K
Bike	7.99M	4.97M	24.93	228K
E-Bike	8.18M	5.05M	25.15	231K



OLA
Overall
Vehicle Type
Revenue
Cancellation
Ratings
Booking Status Breakdown
20M
15M
10M
5M
0M
Value
Cash
UPI
Credit Card
Debit Card
Payment Method
DATE
01-07-2024
31-07-2024
TOP 5 CUSTOMERS
CNR1694595670
CNR3386168975
CNR3706593104
CNR4371152749
CNR5258682402
CNR6153342532
RIDE DISTANCE PER DAY
0.05M
0.04M
0.03M
0.02M
0.01M
0.00M
RIDE DISTANCE
0
5
10
15
20
25
30
35
DATE



OLA
Overall
Vehicle Type
Revenue
Cancellation
Ratings
DATE
01-07-2024
31-07-2024
Total Bookings
103024
Succeeded Bookings
63967
Canceled Bookings
28933
Canceled By Customers
1.01K (9.57%)
1.57K (14.93%)
2.08K (19.82%)
2.67K (25.43%)
3.18K (30.24%)
canceled_rides_by_customer
Driver is not moving towards pickup loca...
Driver asked to cancel
Change of plans
AC is Not working
Wrong Address
Canceled By Rider
2.83K (15.32%)
3.65K (19.82%)
5.41K (29.36%)
6.54K (35.49%)
canceled_rides_by_driver
Personal & Car related issue
Customer related issue
Customer was coughing/sick
More than permitted people in there



OLA
Overall
Vehicle Type
Revenue
Cancellation
Ratings
DATE
01-07-2024
31-07-2024
CUTOMER RATINGS
Prime Sedan
Prime SUV
Prime Plus
Mini
Auto
Bike
E-Bike
4.00
4.00
4.00
4.00
4.00
4.00
3.99
DRIVER RATINGS
Prime Sedan
Prime SUV
Prime Plus
Mini
Auto
Bike
E-Bike
3.99
4.00
4.00
4.00
4.00
3.99
4.00